UNIVERSITY CAMPUS SECURITY MANAGE SYSTEM

PHASE THREE

SOFTWARE PROJECT MANAGEMENT PLAN

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INTRODUCTION

The University Campus Security Management System we are going to create is representative of an upgraded critical infrastructure to strengthen the overall operations and safety of the campus. Within our beginning or initial phase, we select three main categories: the general users(faculty/staff), Security personnel, and the administrative staff as an essential base with different restrictions and powers for the overall system. Ultimately the project adapts an agile process model, allowing for a continuation of interactions between various departments and stakeholders' feedback while maintaining core security requirements.

With a high level of interaction and communication between different bodies of departments including security, students, staff, administrators, faculty, etc. , there is a realization of how many different requirements each group needs, due to the fact that they have such different and polarizing tasks. The constant meetings utilized in the scrum framework align each task in a set order through the backlogs and also allow for conversations to help find solutions that benefit the overall cycle of the University, since there should be no topic of discussion either too big or too small for it not to be included in them. The cycles should be a progressive overload of small steps to reach each department's needs.

The general functionality elaborates on important areas of the security system including CCTV management, incident management, access control, and the emergency alert system. These systems are intended to work in sync, providing a fast and well-managed execution whenever a problem arises. These requirements should highlight how each user will be measured in the intended department, position, location, and restriction. These necessities should equal a user-friendly interface, ensuring it would be accessible for all users on campus while maintaining a high-quality service of security.

SECTION 1: PROCESS MODEL

For the University Campus Security Management System, the agile Scrum framework was chosen as the best process model for this project. Scrum emphasizes iterative development through sprints, which aligns with the nature of this continuously changing system. A campus security system needs to adapt with the changing user needs and emerging threats, and with the flexibility of scrum's process model it is easy to do in short increments. For example, as the security requirements evolve, whether it's updating access control systems, integrating new surveillance technology, or improving emergency alert systems, scrum's rapid adjustments which are done in sprints, ensures the project stays on schedule and current without a massive overhaul of the whole system. This approach that we took takes inspiration heavily from Software Engineering: A Practitioner's Approach (8th ed.), which emphasizes iterative development and adaptability while creating a dynamic system. We made sure to adopt their guidelines on requirements engineering and the agile scrum methodology.

A core aspect of Scrum is emphasis on stakeholder collaboration, which is essential to this project. Security personnel, university administrators, and IT staff are key stakeholders whose requirements will change over time as they deal with emerging threats and technologies. The Scrum framework allows these stakeholders to participate in sprint reviews, providing regular feedback on the upgraded functionalities developed in each sprint. This ensures that the features being built, such as real-time alert systems, or CCTV integration are aligned with the stakeholders immediate needs. For instance, if the university requires more granular access control for sensitive buildings, this feedback can be incorporated into the next sprint, upgrading the system incrementally.

Scrum is also a great fit due to its ability to prioritize certain features through the product backlog, allowing the team to deliver higher value features in a quicker timeline. Given that the project includes mission critical elements like encryption protocols, user restrictions, and mobile notifications, the team can prioritize these security features in the early sprints. This allows key functionalities to be operational as soon as possible while leaving secondary features that can be done in later sprints. This methodology ensures that the system evolves in response to both stakeholder feedback and ongoing security challenges, making Scrum ideal for this complex, and high stakes campus security

management system. While making sure that these features are taken care of early on in the cycle, it is important to make sure that we incorporate security through each phase. This ideal was taken from Baca & Carisson's work on Agile Development with Security Engineering, and will help with the overall integrity of the system.

SECTION II: REQUIREMENTS GATHERING

To gather requirements for the university campus security system, we used different methods. Each method was selected to achieve specific goals and understand user needs thoroughly. First, we interviewed key people such as faculty, students, security staff, and campus officials. These structured interviews provided in-depth insights into their daily tasks, security concerns, and expectations for the system. By talking directly with these stakeholders, we addressed their worries and identified their needs.

Next, we distributed surveys and questionnaires to a larger group of instructors, staff, and students. These tools helped us collect data on safety perceptions, how incidents are reported, and common concerns. The surveys revealed trends and issues that often went unreported, enabling us to prioritize key features like emergency alerts and CCTV monitoring.

We also conducted observations and walkthroughs to see the real-world challenges faced by security personnel. This included watching routine tasks such as patrols, managing access control, and handling incidents. Observing in real time helped us identify practical inefficiencies that might not have come up in interviews or surveys, providing insights into areas like access control and monitoring gaps.

Finally, we analyzed documents by reviewing past incident reports, current security protocols, and university policies. This analysis helped us understand the existing system, identify regulatory requirements, and ensure compliance with university policies. By reviewing these documents, we laid a strong foundation for designing a system that aligns with current protocols while improving to meet future needs.

By using these methods together, we gained a complete understanding of the requirements for the campus security system, ensuring the final design meets user needs, functions effectively, and follows regulations.

RESULTS

- **Interviews:**
  - **Stakeholders stressed the need for a fast and easy incident reporting system.**
  - **Security staff focused on the importance of real-time CCTV monitoring and improved access control features.**

- **Surveys:**
  - **60% of participants felt that some campus areas were unsafe at night.**
  - **65% said they did not report minor incidents because the process took too long.**
  - **The top requested features included emergency alerts (87%) and better surveillance (73%).**

- **Observations:**
  - **We found issues in how patrols were scheduled and how access control was managed.**
  - **There is a need for better integration between security procedures and real-time monitoring tools.**

- **Document Analysis:**
  - **We discovered inconsistencies in past incident reporting and gaps in policy enforcement.**
  - **We confirmed that new systems must meet legal and institutional requirements.**

Figures:

SECTION III: REQUIREMENTS

Functional Requirements

❖ **Incident Reporting System**

- Enable users to report incidents quickly and easily.
- Organize incidents by type and urgency.
- Allow users to attach photos or videos as evidence.

❖ **Real-Time Monitoring**

- Provide live access to CCTV feeds with user-friendly controls.
- Send alerts for unusual activity or unauthorized access.

❖ **Emergency Warning System**

- Distribute emergency alerts via SMS, email, and app notifications.
- Provide clear instructions for safety procedures or evacuation.

❖ **Access Control Management**

- Regulate who can enter restricted areas.
- Maintain records of who accessed specific areas and when.

❖ **Surveys and Feedback Collection**

- Create a platform for users to share feedback on safety concerns.
- Allow anonymous submissions to encourage honest input.

❖ **Data Visualization**

- Offer dashboards to track safety trends and identify problem areas.

- Utilize visual tools like heat maps to highlight unsafe locations.

Non-Functional Requirements

❖ **Ease of Use**

- Design the system to be simple for all users, including faculty, students, and staff.

- Provide support for multiple languages.

❖ **Performance**

- Ensure real-time updates for incident reporting and CCTV feeds.

- Support a high volume of users without performance degradation.

❖ **Scalability**

- Develop the system to accommodate growth as more users or locations are added.

- Allow for the integration of new features, such as AI-based tools, in the future.

❖ **Reliability**

- Aim for high system uptime (99.9%).

- Include data backup and recovery options.

❖ **Compliance and Security**

- Adhere to all relevant regulations and privacy laws.

- Safeguard sensitive data and perform regular security checks.

❖ **Customizability**

- Enable adjustments to meet the specific needs of different user groups.

- Allow users to customize settings based on their roles.

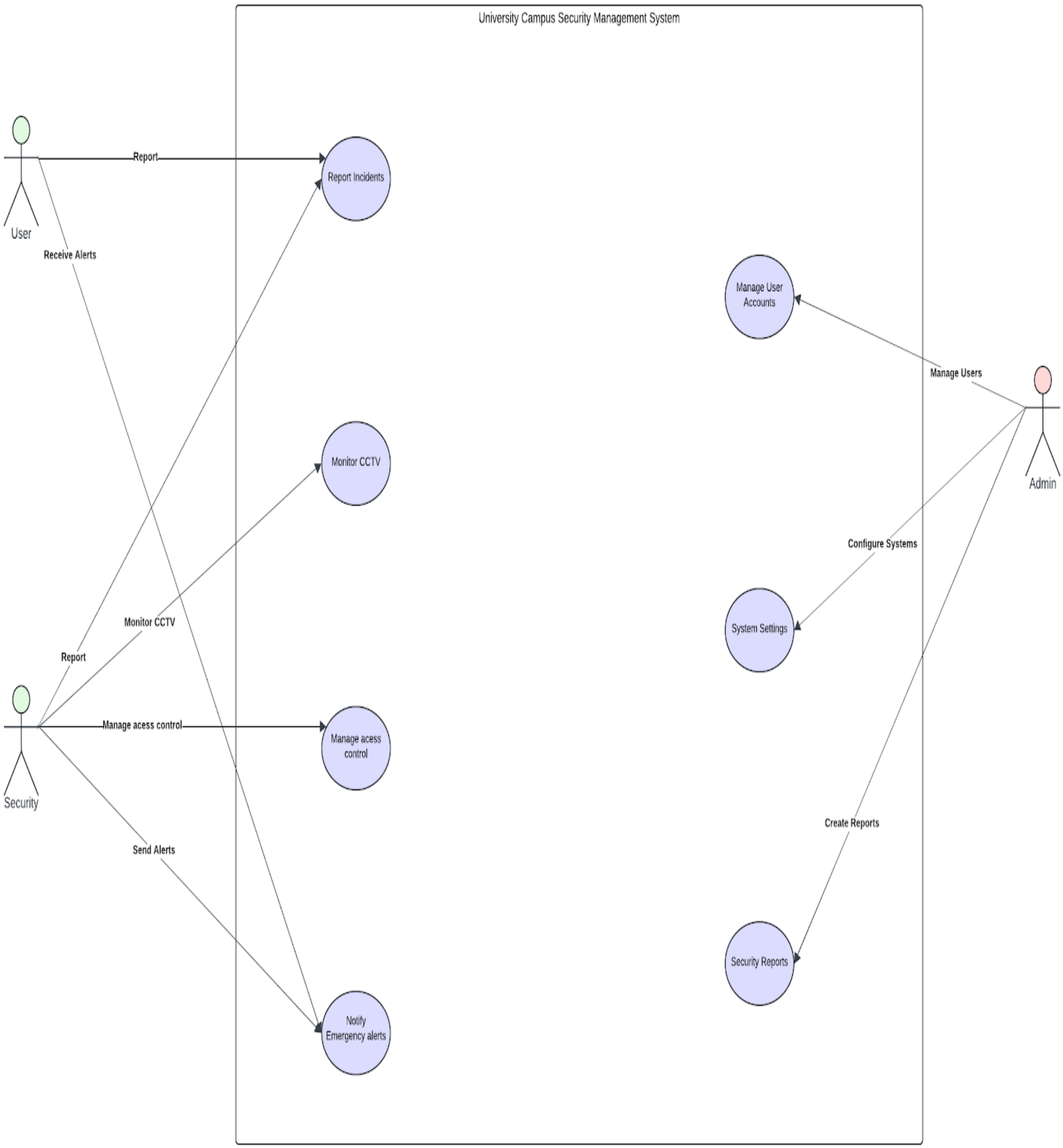
❖ **Integration**

- Ensure smooth operation with existing campus systems, such as ID card access.

- Facilitate connections with other tools via APIs to share and utilize data effectively.

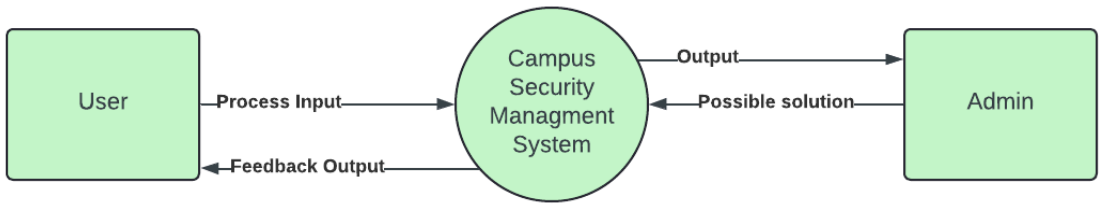
SECTION IV: REQUIREMENTS SPECIFICATIONS

Case Diagram

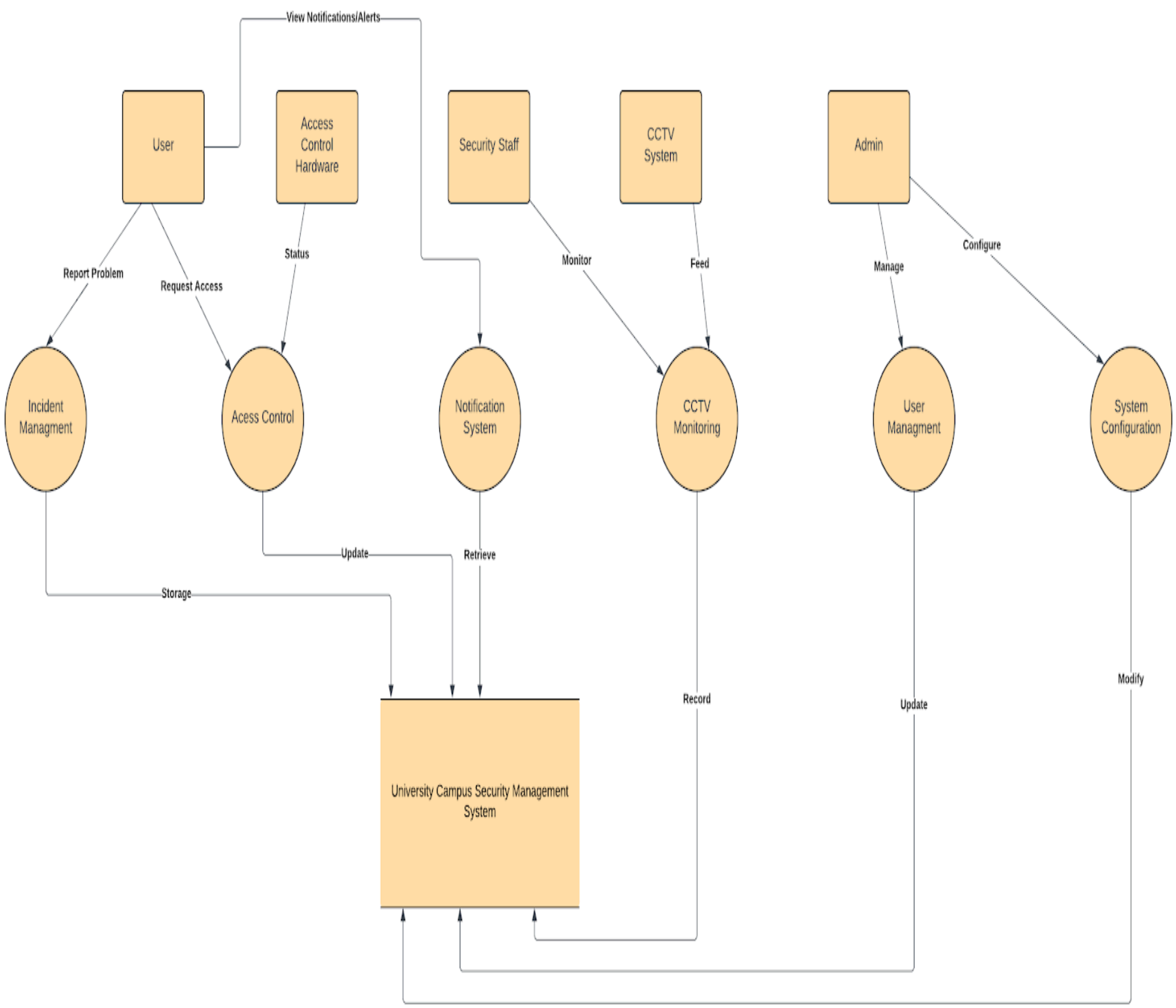


SECTION V: DOMAIN ANALYSIS (DATA FLOW DIAGRAM)

Level 0 DFD



Level 1 DFD



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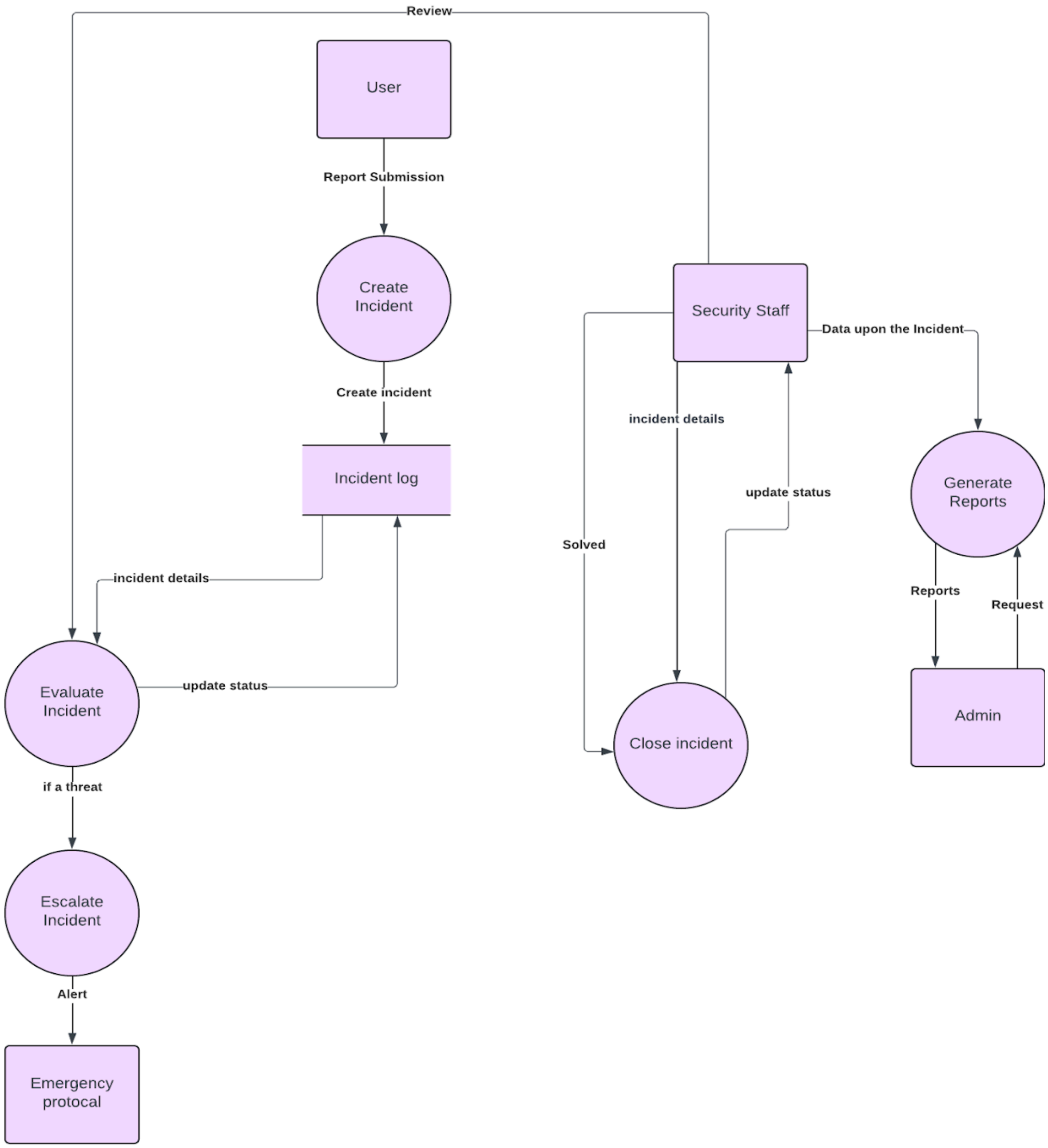
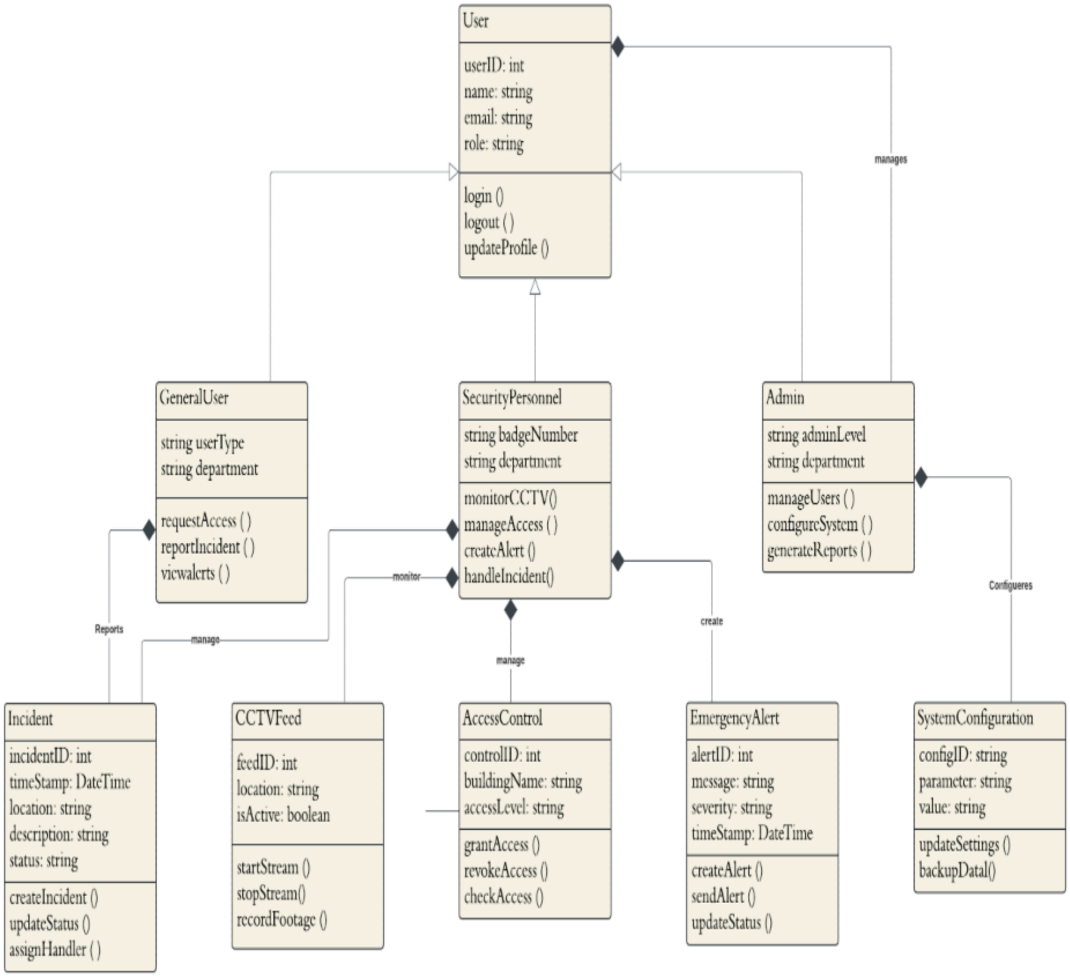
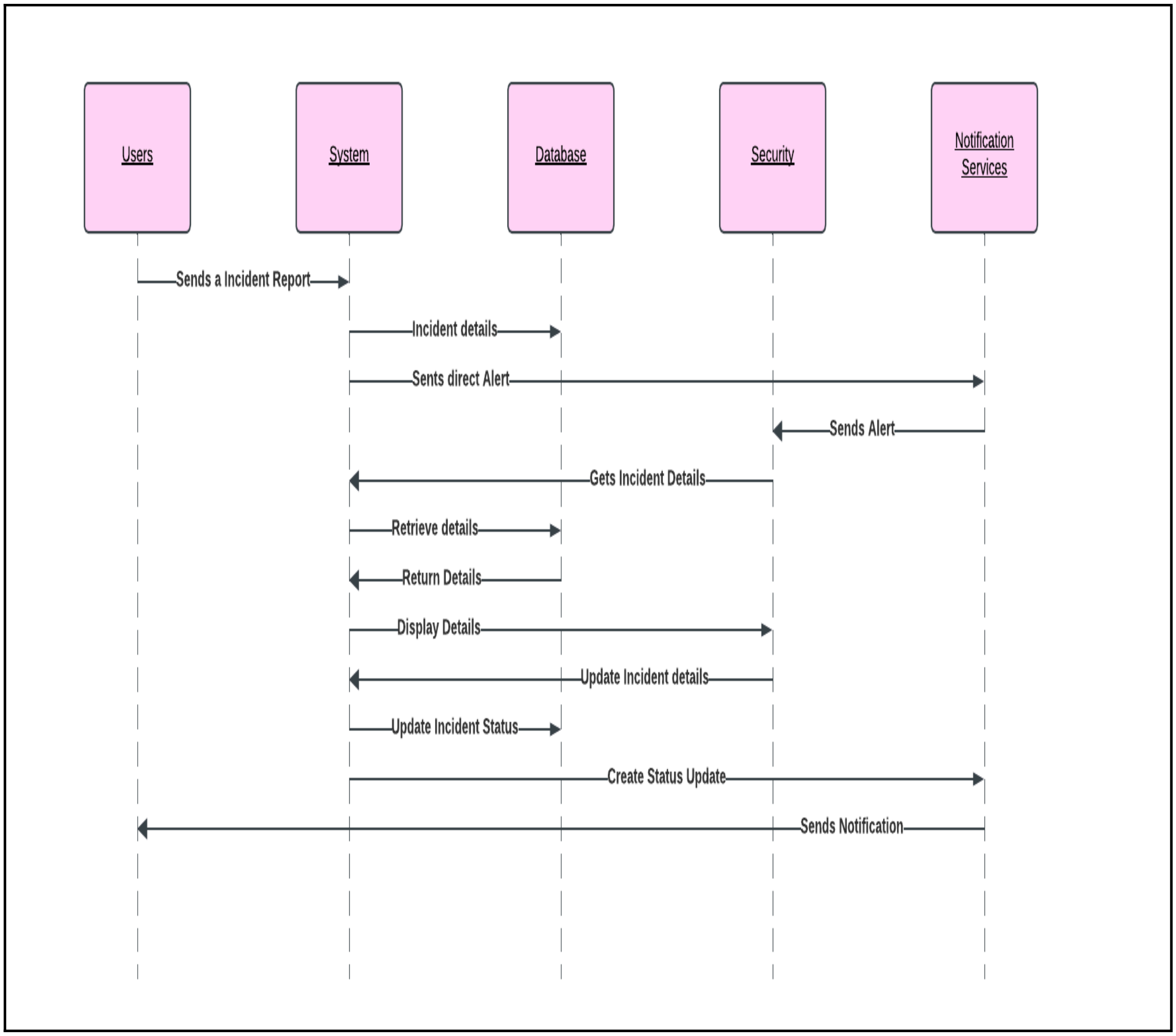

Review
User
Report Submission
Create Incident
Create incident
Incident log
incident details
update status
Evaluate Incident
if a threat
Escalate Incident
Alert
Emergency protocal
Security Staff
Data upon the Incident
incident details
update status
Solved
Close incident
Generate Reports
Reports
Request
Admin

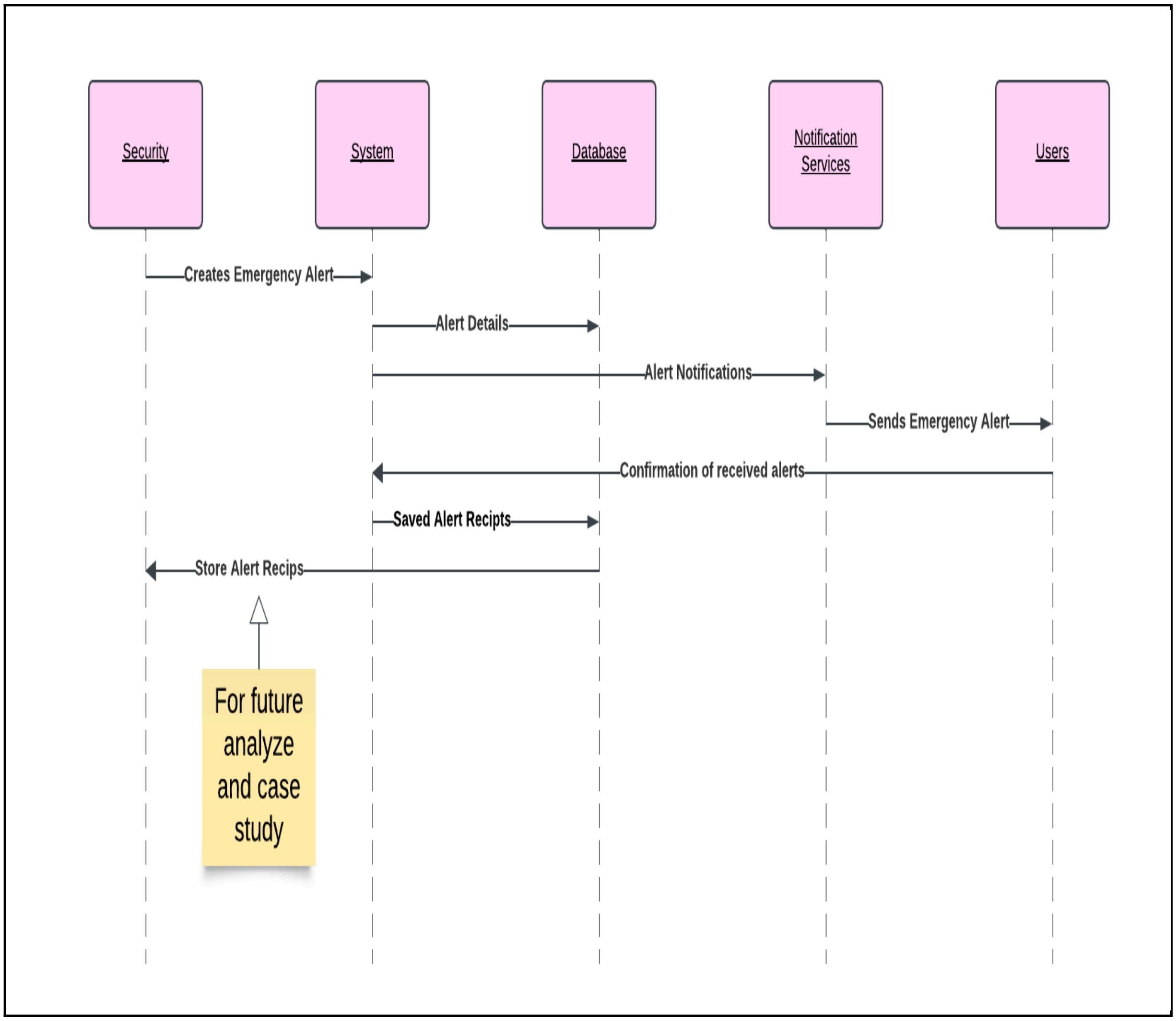
SECTION VI: CLASS MODEL



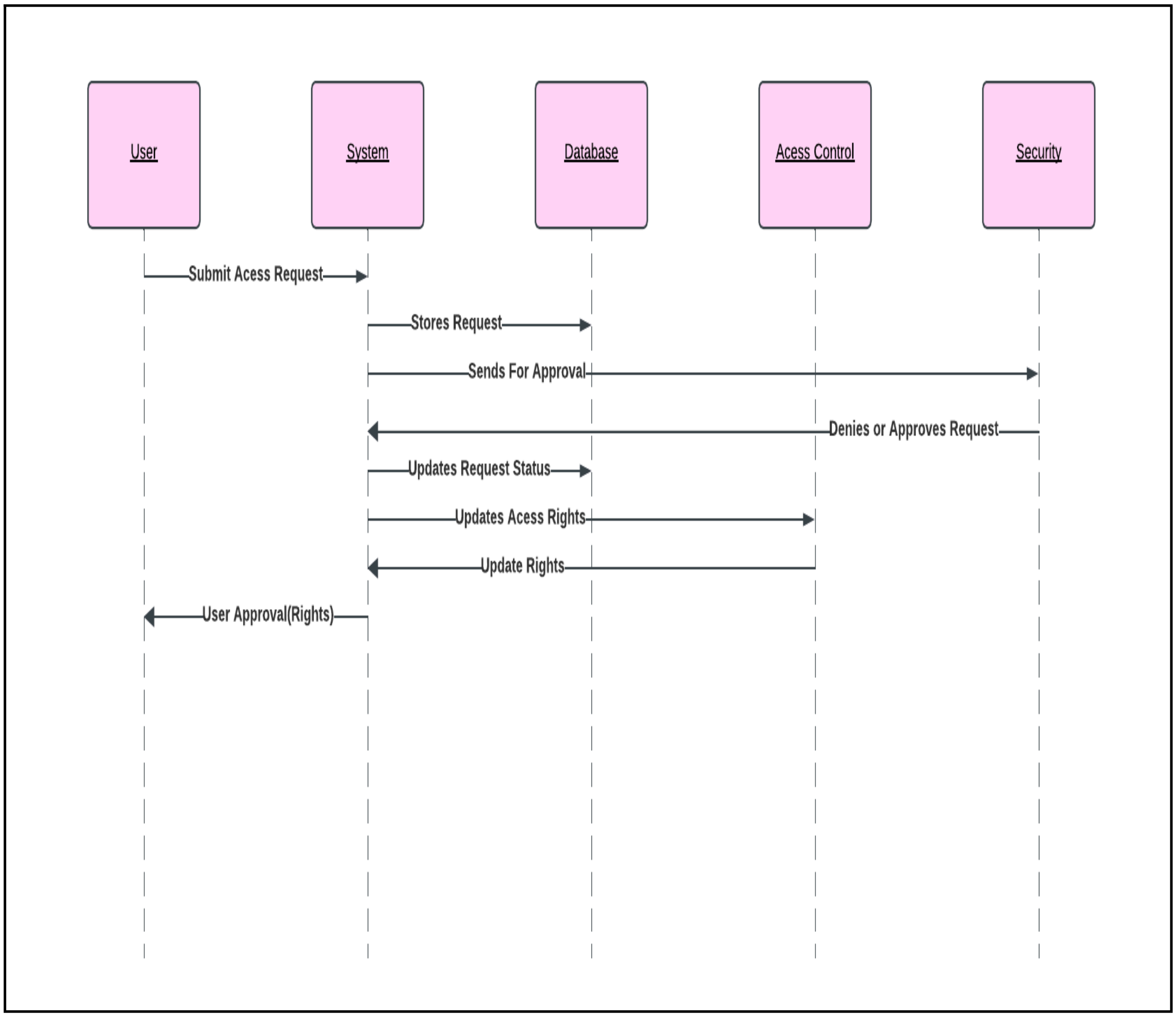
SECTION VII: SEQUENCE MODEL

Use Case 1: Incident Reporting
For the incident reporting the use case diagram represents how the functions of all personal including users(students/faculty) can report any incident. When any occurrence emerges, the user reports or initiates the reporting process to the system interface. The incident should be concluded once the user fully completes the reports; so probable importance features must be filled to report such as description, location, and the importance or severity of the situation. System interface then stores the data in the database and creates a notification toward the security personal. The security personal now can view the report in the system interface and update the database. With continuation to update the overall status this shows a demonstration to react fast and efficiently to security concerns, while achieving to keep all parties informed on the status and actions taken to any respective situation.



Use Case 2: Emergency Alert
The Emergency Alert System use case is very effective focus on the distribution across campus. With process being initiated only when the security department view any event as immediate importance to send throughout the whole campus' notifications systems. First the security will have to create an alert description on severity, affected parts, and a detailed messaged on the manner of events. The message then is stored in the database and processes to the notification system and messages all user within the multiple distributions of notification including email, text, phone calls, etc. (Depending either on the user's best form of method to communicate or the employees' setting of notifications). A verification of successful messages should later retrieve toward the systems interface where log sheets should be saved in the database and also stored within the department to study for future reference on either how to be more effective or better their services.


Use Case 3: Access Control Request
This section would give a glimpse of how the workers either maintain or manage the areas across campus. Specifically, when users from all departments need access to certain areas. With a time stamp of either entering and leaving this specific locations that require higher entrance credentials. The sequence begins with the user entering his credentials, or scanning their ID. Their Access info is sent straight toward the system interface and saved in the database. Once then security could either approve or analyze the situation based of credentials, universities policies, and security protocols. Then approved it gives confirmation toward the access controls and also keeps a log in the system interface. Which then finally grants full access to the user, justifying such a process to keep access control interfaces an important priority keeping the campus safe of all physical and cyber attacks.



SECTION VIII: COMPLETED AND PLANNED ACTIVITIES

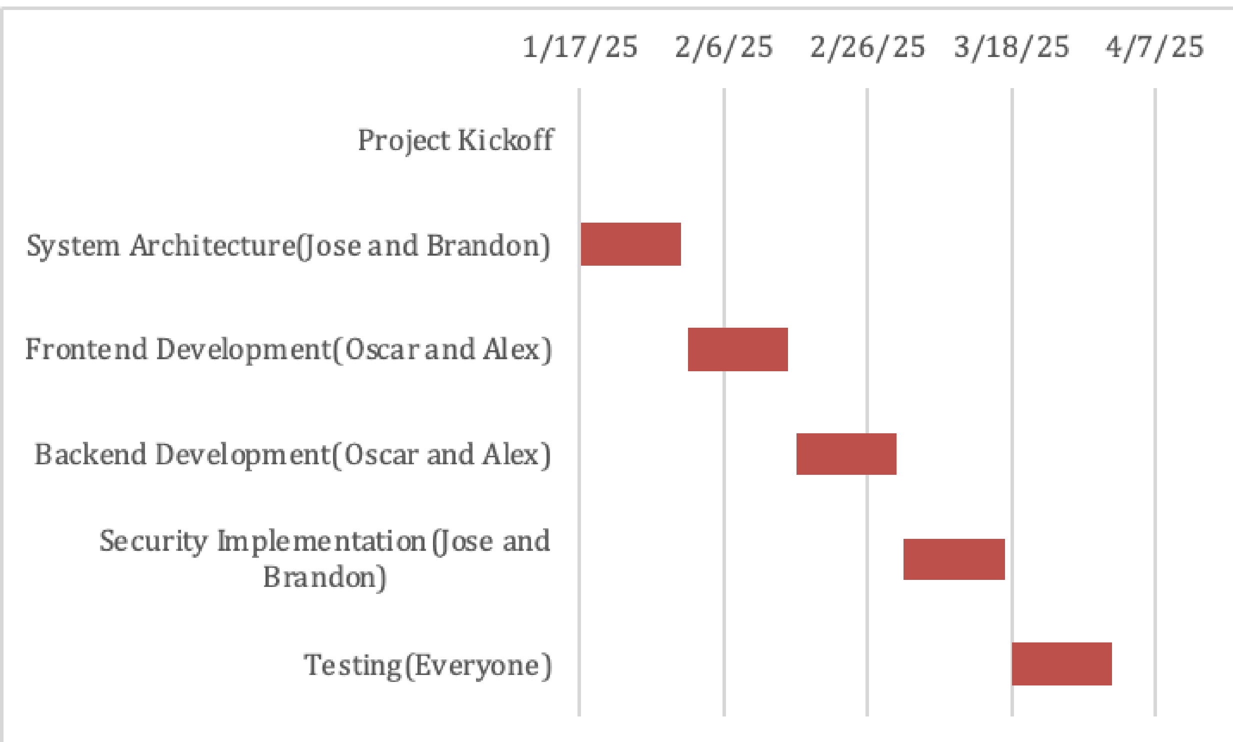
Previous semester

Name	Task
Alex	**Managing project:** • **Conducted initial project gathering and scoping** • **Developed project management plan** • **Started introduction** • **Process models**
Oscar	**Software architecture:** • **Design the class diagrams** • **Defined system components and interactions** • **Created the initial technological architectures** • **Evaluated all potential technologies and frameworks** • **Create the class diagrams and sequence diagrams**
Jose	**System Analyst:** • **Performed a detailed analysis of the requirements** • **Created use case and sequence diagrams** • **Create initial system architecture** • **Documented several system requirements and their specifications**
Brandon	**Backend:** • **Analyzed several system requirements for backend implementation** • **Researched backend technologies** • **Developed the initial data models** • **Created preliminary system flow diagrams to ease the process.**

Next Semester

Alex	**Managing project:** ● **Oversee the system implementation needed for progression** ● **Manage project resources** ● **Prepare final project documentation** ● **Manage project milestones and the deadlines**
Oscar	**s.**
Jose	**System Analyst:** ● **Refine the system design based on implementation feedback from constant interaction and critics.** ● **Assist in system integration throughout the semester.** ● **Develop comprehensive test cases**
Brandon	**Backend:** ● **Implement the core backend services.** ● **Implement authorization and authentication for all users and admin.**

Task	Start	End	Duration
Project Kickoff	1/17/25	1/17/25	0
System Architecture(Jose and Brandon)	1/17/25	1/31/25	14
Frontend Development(Oscar and Alex)	2/1/25	2/15/25	14
Backend Development(Oscar and Alex)	2/16/25	3/2/25	14
Security Implementation(Jose and Brandon)	3/3/25	3/17/25	14
Testing(Everyone)	3/18/25	4/1/25	14



CONCLUSION

In this 3rd Phase we offered a complete design of the university campus security management system, making sure to highlight its distinctive functionality and integration of advanced security measures. Our use case's requirements are reflected in the class diagram, which emphasizes access control, CCTV integration, incident management, and emergency notifications. These elements provide a thorough approach to campus safety and security while leveraging the agile principles, enabling us to maintain a scalable and easily maintained system while ensuring that stakeholder needs are satisfied.

We also placed a lot of emphasis on security measures, which were informed by IBM's 2023 Cost of a Data Breach Report, which highlights the significance of taking preventative action to reduce risks. It had a direct impact on our implementation of dynamic monitoring systems and sophisticated access control. We were also motivated to incorporate security considerations at every stage of the system life cycle by Baca & Carisson's work on Agile Development with Security Engineering. The system is fully equipped to handle new threats by incorporating security into all stages of its life cycle. Our solution is comprehensive and feasible for a campus-wide installation thanks to the study that was done to determine what we should accomplish for the project.

After the semester is over and we start the development process of this project our team will focus on iterative development, consistent with Agile methodology discussed by Pressman and Maxim. This will involve breaking implementation into sprints, allowing continuous improvement and feedback from stakeholders. As well we will focus on rigorous testing, both functional and security based, to validate the system' performance under real world conditions. Deployment of the system will begin with a phase where we gather user feedback in order to further improve the system. The last part will be giving user training and awareness of how the programs work to ensure the system is used effectively, improving campus wide security practices.

This project demonstrates how collaboration between stakeholders, and research of already available practices, can create a comprehensive security solution. By placing a strong emphasis on adaptability, we can make sure that our system changes to meet the demands of both the institution where it is implemented and its users. The project has a good chance of success by investigating popular software engineering concepts and state of the art security procedures. By including the robust security measures mentioned in the works and applying the agile approaches that have been studied, the system will achieve the objectives of accessibility and safety. The University Campus Security Management System is a dependable and efficient solution for any contemporary campus setting thanks to all the actions we performed.

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APPENDIX

Provide all material used for surveys and requirements gathering.

Student Survey:

1. Do you feel safe on campus during the day?(Yes/No)

2. Do you feel safe on campus during the day?(Yes/No)

3. Are you aware of how to report incidents currently on campus? (Yes/No)

4. Have you ever reported an incident on campus? (Yes/No)

5. Which features would improve your sense of safety on campus? (Check all that applies)

- **Emergency alert notifications**
- **Improved CCTV coverage**
- **Access control for campus buildings**
- **More security personnel**

6. Do you think real time monitoring and incident reporting through an app would be useful? (Yes/No)

25 Students Surveyed on campus:

- **20 Students out of 25 students feel safe on campus during the day, but 15 out of 25 feel safe at night**
- **20 out of 25 students were unaware of how to report incidents on campus**
- **The most requested features were emergency alert notifications and improved CCTV coverage**